CHAPTER 4

Random Variables

4.1 Introduction

Suppose we have a sample space S and a probability function $\mathbb{P}$ which assigns a number (between 0 and 1) to each event. In order to be able to manipulate the events on the sample space mathematically, it is necessary that a numerical value is attached to each event. For example, in tossing 3 coins, one might be interested in the number of heads obtained and not really concerned about individual outcome of each coin. In that case, we would have

$$\begin{array}{rcccccccc} S = & \{HHH & HHT & HTH & THH & HTT & THT & TTH & TTT\} \\ X = & 3 & 2 & 2 & 2 & 1 & 1 & 1 & 0 \end{array}$$

Thus the event '$X = 1$' corresponds to the subset $\{HTT, THT, TTH\}$ of S and therefore $\mathbb{P}(X = 1) = 3/8$.

Once all the elementary events in a sample space have been assigned numerical values, we let X 'stand for' the numerical values of the elementary events. We then call X a *random variable*. We refer to X as random variable for the following reasons:

- *random* because its values depend on the outcome of a random experiment;

- *variable* because it takes on several values.

Definition 4.1.1. A *random variable* is a function whose domain is a sample space and whose range is the real line, i.e.

$$X : S \to \mathbb{R} \tag{4.1}$$

Expressed somewhat differently, a random variable X is a numerical value whose value is determined by the outcome of a random experiment. We usually use uppercase letters to denote different random variables but the most common one is X.

Example 4.1.1. Suppose you have 5 coins in a pocket – two P1 coins, two P2 coins and P5 coin – and you pull out two coins at random for a tip. Let the random variable X be the amount of the tip. What are the possible values that X can take and their corresponding probabilities?

Solution 4.1.1. Denote the different coins as follows: $1_1, 1_2, 2_1, 2_2$ and 5. Then the sample space and the numerical values assigned to each event are as follows:

$$\begin{array}{lcccccccccc} S = & \{1_1 1_2 & 1_1 2_1 & 1_1 2_2 & 1_1 5 & 1_2 2_1 & 1_2 2_2 & 1_2 5 & 2_1 2_2 & 2_1 5 & 2_2 5\} \\ X = & 2 & 3 & 3 & 6 & 3 & 3 & 6 & 4 & 7 & 7 \end{array}$$

Thus, the values that X can take and their probabilities are

$X = x$	2	3	4	6	7
$\mathbb{P}(X = x)$	0.1	0.4	0.1	0.2	0.2

Example 4.1.2. A car salesperson is scheduled to see two clients today. He sells only two models of cars, an `executive` (E) and a `basic` (B) models. Each executive model sold earns the salesperson a commission of P2000.00 while each basic model sold earns

the person only P1000.00. If the sale is `lost` (L) no commission is earned. Suppose $\mathbb{P}(E) = 0.2$, $\mathbb{P}(B) = 0.3$ and $\mathbb{P}(L) = 0.5$. Also assume that the sales are independent of each other. Let the random variable X be the total commission earned by the salesperson today. What values can X take and with what probabilities?

Solution 4.1.2. It should be noted that unlike the previous example, here the order matters. For example, BL would imply that the first buyer bought a basic model while the second buyer did not buy any of the models. Therefore the sample space of the salesperson and the corresponding commission made are given by

$$\begin{array}{lcccccccccc} S = & \{ & LL & LB & LE & BL & BB & BE & EL & EB & EE \ \} \\ X = & & 0 & 1000 & 2000 & 1000 & 2000 & 3000 & 2000 & 3000 & 4000 \end{array}$$

Thus, the values that X can take and their probabilities are computed as follows:

$$\begin{aligned} \mathbb{P}(X = 0) &= \mathbb{P}(LL) = \mathbb{P}(L)\mathbb{P}(L) \qquad \text{(by independence)} \\ &= 0.5(0.5) = 0.25; \\ \mathbb{P}(X = 1000) &= \mathbb{P}(LB \text{ or } BL) = 2\mathbb{P}(LB) \\ &= 2(0.5)(0.3) = 0.30; \\ \mathbb{P}(X = 2000) &= \mathbb{P}(BB \text{ or } LE \text{ or } EL) \\ &= (0.3)(0.3) + 2(0.5)(0.2) \\ &= 0.29. \end{aligned}$$

Similarly $\mathbb{P}(X = 3000) = 0.12$ and $\mathbb{P}(X = 4000) = 0.04$. Hence

$X = x$	0	1000	2000	3000	4000
$\mathbb{P}(X = x)$	0.25	0.30	0.29	0.12	0.04

4.2 Discrete and Continuous Random Variables

Random variables can be categorized into two groups – *discrete* or *continuous*. The mathematical treatment of these two types of random variables is very different. We give a brief description of the two types and also given examples of random variables for each group below.

Discrete random variables: take on isolated values along the real line, usually integer values but by no means always. The following are some examples of integer-valued discrete random variables:

- the number of customers entering a store between 17h00 and 18h00;
- the number of occupied tables in a restaurant;
- the number of shots on target before the fist goal of the home team.

Continuous random variables: can (conceptually, at least) be measured to any degree of accuracy. That is, between every two possible values x_1 and x_2 that a random variable can assume, there is another possible value x such that $x_1 \leq x \leq x_2$. The set of all possible values of a continuous random variable is usually an interval on the real line. Examples of continuous random variables are:

- the distance traveled by a car in one litre of petrol;
- the proportion of gold in a sample of ore;
- the time that a customer waits in any queue;
- the volume of milk that actually goes into a nominally one litre carton.

Example 4.2.1. State whether the following are random variables, if they are, which ones are continuous or which ones are discrete? Write down the set of values that each random variable can take on.

(a) The number of customers arriving at a supermarket during the lunch hour.

(b) The number of letters in the Greek alphabet.

(c) The opening price of gold ETF at Botswana Stock Exchange on a Friday.

(d) The ratio between the circumference and the diameter of a circle.

(e) The last digit of a randomly selected cellphone number.

Solution 4.2.1. Left as an exercise!

4.3 Distribution Functions

4.3.1 Probability Mass Function

For a discrete random variable X, we define the probability mass function, $p(x)$ of X by

$$p(x) = \mathbb{P}(X = x) \tag{4.2}$$

The probability mass function (p.m.f) should satisfy the following conditions:

(a) $p(x)$ is defined for all values of x but $p(x) \neq 0$ only at a finite or countably infinite set of points

(b) $0 \leq p(x) \leq 1$

(c) $\sum_x p(x) = 1$

We now consider several examples of probability mass functions

Example 4.3.1. An unbiased die is rolled and the random variable X consist of the number of dots appearing on the upturned face. Find the probability mass function for this random variable .

Solution 4.3.1. simply letting $\mathbb{P}(X = x) = p(x)$ gives us the the required p.m.f. because the die is unbiased

$$\mathbb{P}(X = 1) = p(1) = \frac{1}{6}$$

and similarly

$$p(2) = p(3) = \ldots = p(6) = \frac{1}{6}.$$

All other values of X represent impossible events.

Example 4.3.2. The Minister of Environment, Wildlife & Tourism has to decide on fishing quota for the forthcoming season. Currently, the biomass of fish is estimated to be 20 000 tonnes. The fish may have a good breeding season with probability `0.3` and produce 10 000 tonnes of young fish, or have a bad breeding season and produce only 1 000 tonnes. A so-called 'warm-water event' may occur with probability `0.1` and kill 15 000 tonnes of fish, otherwise 1 000 tonnes of fish will die naturally.

Find the p.m.f for X, the biomass of fish before setting the quota (assuming all events are independent). If the Minister bases his decision using a policy that the biomass must remain 10 000 tonnes or more with probability `0.8`, what should his decision be?

Solution 4.3.2. Let G represent good breeding seasons W represent the occurrence of the warm-water event. Then from the statements above, we have that:

$$\mathbb{P}(G) = 0.3 \qquad \& \qquad \mathbb{P}(W) = 0.1.$$

There are four possible scenarios that can happen in the forthcoming season – good season & no warm-water event, good season & warm-water event, bad season & no-warm water event or bad season & warm-water event. Thus, the sample space is given by

$$S = \left\{ G\overline{W}, \;\; GW, \;\; \overline{G}\,\overline{W}, \;\; \overline{G}W \right\}$$

Now consider the random X, which represent the biomass of fish under each possible scenario given that the current biomass is 20 000 tonnes.

i. For event $G\overline{W}$ we have

$$\begin{aligned} X &= 20000 + 10000 - 1000 \\ &= 29000 \end{aligned}$$

and it follows that

$$\begin{aligned} \mathbb{P}(X = 29000) &= \mathbb{P}(G\overline{W}) \\ &= \mathbb{P}(G)\mathbb{P}(\overline{W}) \qquad \text{(independence)} \\ &= 0.3(1 - 0.1) = 0.27. \end{aligned}$$

ii. Similarly for the event GW:

$$\begin{aligned} X &= 20000 + 10000 - 15000 \\ &= 15000 \end{aligned}$$

and it follows that

$$\begin{aligned} \mathbb{P}(X = 15000) &= \mathbb{P}(GW) \\ &= \mathbb{P}(G)\mathbb{P}(W) \\ &= 0.3(0.1) = 0.03. \end{aligned}$$

iii. For the event $\overline{G}\,\overline{W}$ (bad season & no warm-water event):

$$\begin{aligned} X &= 20000 + 1000 - 1000 \\ &= 20000 \end{aligned}$$

and it follows that

$$\begin{aligned} \mathbb{P}(X = 20000) &= \mathbb{P}(\overline{G}\,\overline{W}) \\ &= \mathbb{P}(\overline{G})\mathbb{P}(\overline{W}) \\ &= (1 - 0.3)(1 - 0.1) = 0.63. \end{aligned}$$

iv. Lastly, for the event $\overline{G}\,W$ (bad season & warm-water event):

$$\begin{aligned} X &= 20000 + 1000 - 15000 \\ &= 6000 \end{aligned}$$

and it follows that

$$\begin{aligned} \mathbb{P}(X = 20000) &= \mathbb{P}(\overline{G}\,W) \\ &= \mathbb{P}(\overline{G})\mathbb{P}(W) \\ &= (1 - 0.3)(0.1) = 0.07. \end{aligned}$$

Hence the probability mass function for X is given by:

$X = x$	6	15	20	29
$p(x)$	0.07	0.03	0.63	0.27

From this p.m.f $\mathbb{P}(X \geq 10000) = 0.93$, therefore the Minister should set the quota at 10 000 tonnes in biomass of fish.

For a discrete random variable X, the probability of the event $a \le X \le b$ is found by summing all probabilities of the values of X that lie between a and b, a & b inclusive. That is,

$$\mathbb{P}(a \le X \le b) = \sum_{x=a}^{b} p(x) \tag{4.3}$$

BE CAREFUL WHEN HANDLING '$\le$' & '$<$' OR '$\ge$' & '$>$'!!!

Example 4.3.3. Let

$$p(x) = \begin{cases} x/15 & x = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

(a) Check that the function $p(x)$ satisfies the conditions for being the probability mass function for some random variable X.

(b) Find $\mathbb{P}(2 \le X \le 4)$.

(c) Compute $\mathbb{P}(X > 3)$.

Solution 4.3.3. These are pretty straight forward and you should be able to do them with minimal effort. But in any case, here are the solutions.

(a) We are required to show that for all values of X, $p(x)$ is non-negative and their sum adds up to 1. We have that

$$p(1) = 1/15;\ p(2) = 2/15;\ p(3) = 3/15;\ p(4) = 4/15;\ p(5) = 5/15$$

and all these fall between 0 and 1. Their sum is

$$\sum_{x=1}^{5} p(x) = (1 + 2 + 3 + 4 + 5)/5 = 1.$$

Hence $p(x)$ is a probability mass function.

(b) We have that

$$\begin{aligned}\mathbb{P}(2 \leq X \leq 4) &= p(2) + p(3) + p(4) \\ &= \frac{(2+3+4)}{15} \\ &= 0.6.\end{aligned}$$

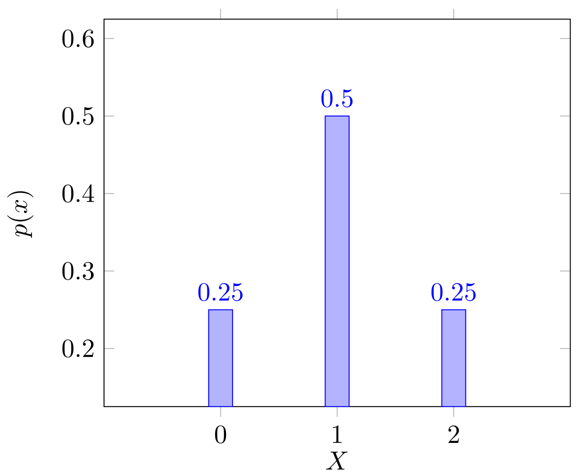
(c) You should take note of the strict inequality here.

$$\begin{aligned}\mathbb{P}(X > 3) &= p(4) + p(5) \\ &= \frac{4+5}{15} \\ &= 0.6.\end{aligned}$$

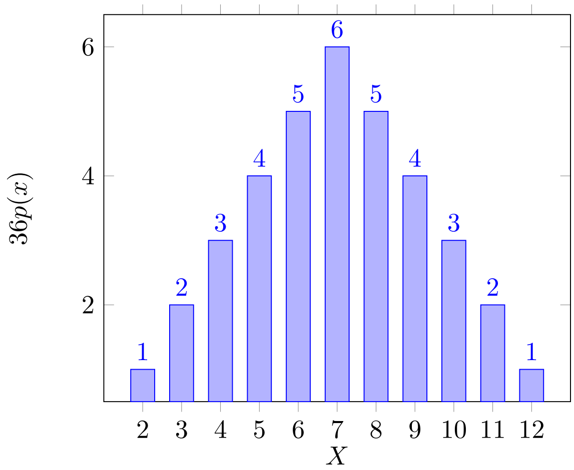
A probability mass function is conveniently plotted by means of a bar graph. This gives an easily interpretable visual impression of the shape of the distribution of probabilities associated with the random variable. For instance, if the probability mass function of X is

$$p(0) = 1/4; \qquad p(1) = 1/2; \qquad p(2) = 1/4,$$

then its bar graph is given as follows.



Similarly, a graph of the probability mass function of the random variable representing the sum when two dice are rolled is presented below.



Example 4.3.4. Let

$$p(x) = \begin{cases} \binom{3}{x} 0.5^x & x = 0, 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Plot a bar graph showing the distribution of the probabilities of $p(x)$.

(b) Show that $p(x)$ is a probability mass function for the random variable X that represent the number of heads obtained when three coins are flipped.

(c) Find $\mathbb{P}(X \leq 2)$.

Solution 4.3.4. The solution left for the reader to attempt as a practice problem.

Cumulative Distribution Function

The cumulative distribution function (c.d.f) of a discrete random variable X, denoted by F, is expressed as follows

$$F(a) = \mathbb{P}(X \leq a) = \sum_{\text{all } x \leq a} p(x). \tag{4.4}$$

Suppose that X takes the values $x_1, x_2, x_3, \ldots$ such that $x_1 < x_2 < x_3 < \ldots$. Then the cumulative distribution (commonly referred to as just the distribution function), F is a step function. That is, the value of F remain constant in the intervals $[x_{i-1}, x_i)$, then takes a step/jump of size $p(x_i)$ at x_i. For example, if X has a probability mass function given by

$$p(1) = 1/4,\ p(2) = 1/2,\ p(3) = 1/8 \text{ and } p(4) = 1/8;$$

then its distribution function is

$$F(x) = \begin{cases} 0 & x < 1 \\ 1/4 & 1 \leq x < 2 \\ 3/4 & 2 \leq x < 3 \\ 7/8 & 3 \leq x < 4 \\ 1 & x \geq 4. \end{cases}$$

Properties of the Cumulative Distribution Function

The following are some of the properties of the cumulative distribution function, F.

(a) F is a non-decreasing function. That is, if $a < b$ then $F(a) \leq F(b)$.

(b) $\lim_{x\to\infty} F(x) = 1$.

(c) $\lim_{x\to-\infty} F(x) = 0$.

(d) F is right continuous. That is, for any a and any decreasing sequence a_n, $n \geq 1$, that converges to a, $\lim_{n\to\infty} = F(b)$.

Example 4.3.5. The distribution function of a random variable X is given by

$$F(x) = \begin{cases} 0 & x < 0 \\ 1/2 & 0 \leq x < 1 \\ 2/3 & 1 \leq x < 2 \\ 11/12 & 2 \leq x < 3 \\ 1 & x \geq 3. \end{cases}$$

Compute the following.

(a) $\mathbb{P}(X < 3)$

(b) $\mathbb{P}(X = 1)$

(c) $\mathbb{P}(X > 0.5)$

(d) $\mathbb{P}(2 < X \leq 4)$

Solution 4.3.5. Below are the solutions.

(a) $\mathbb{P}(X < 3) = F(2) = 11/12$.

(b) $\mathbb{P}(X = 1) = F(1) - F(0) = 2/3 - 1/2 = 1/6$.

(c) $\mathbb{P}(X > 0.5) = 1 - F(0) = 1 - 1/2 = 1/2$.

(d) $\mathbb{P}(2 <\leq 4) = F(4) - F(2) = 1 - 11/12 = 1/12$.

4.3.2 Probability Density Function

Continuous random variables are represented by *probability density function* (p.d.f) which is mathematically treated differently from the probability mass functions. We therefore adopt different notation in order to make a distinction between a density function and mass function. We denote the probability density function by $f(x)$ instead of $p(x)$ used for the probability mass function.

Definition 4.3.1. A function $f(x)$ is called a *probability density function* if it satisfies the following conditions:

(a) $f(x)$ is defined for all values of X.

(b) all values of $f(x)$ are non-negative, i.e. $0 \leq f(x) < \infty$.

(c) $\int_{\infty}^{\infty} f(x)dx = 1$. That is, the 'area under the curve' of the probability density function is one.

We have seen that the probabilities for discrete are found by calculating the values of probability mass function, $p(x)$ at the points of interest and summing them over the required interval. That is,

$$\mathbb{P}(a \leq X \leq b) = p(a) + p(a+1) + \ldots + p(b-1) + p(b).$$

However, for continuous random variables, the probability density function $f(x)$ is constructed in such a way that the probabilities of events are found by integration. That is,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)dx. \tag{4.5}$$

Note that if we let $b = a$, then $\mathbb{P}(a) = \int_a^a f(x)dx = 0$. Thus, the probability that a continuous random variable will assume a fixed value is zero. Hence for continuous random variables, we compute

$$F(a) = \mathbb{P}(X < a) = \mathbb{P}(X \leq a) = \int_{-\infty}^{a} f(x)dx \tag{4.6}$$

which is the distribution function of X.

Example 4.3.6. Let

$$f(x) = \begin{cases} 6x(1-x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that $f(x)$ is a probability density function.

(b) Compute $\mathbb{P}(0.4 \leq X \leq 0.6)$.

Solution 4.3.6. Note that to show that a function is a p.d.f, all we are required to do is show that the integral of $f(x)$ over the interval of X where the probabilities are non-zero is one.

(a) We have

$$\begin{aligned} \int_0^1 f(x)dx &= \int_0^1 6x(1-x)dx \\ &= 6\int_0^1 (x - x^2)dx \\ &= 6\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 \\ &= 6\Big[\big(1/2 - 1/3\big) - \big(0/2 - 0/3\big)\Big] \\ &= 6(1/6) = 1. \end{aligned}$$

(b) From the solution above, we have that

$$\begin{aligned}
\mathbb{P}(0.4 \leq X \leq 0.6) &= \int_{0.4}^{0.6} 6x(1-x)dx \\
&= \\
&\vdots \\
&= 6\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{0.4}^{0.6} \\
&= 0.296.
\end{aligned}$$

Example 4.3.7. Let

$$f(x) = \begin{cases} k(x^2-1) & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the value of k if $f(x)$ is a probability density function.

(b) Compute $\mathbb{P}(2 \leq X \leq 3)$.

Solution 4.3.7. Attempt it on your own.

Example 4.3.8. A remote country service station is supplied with petrol once a week. The weekly demand for petrol (measured in 1000L) is a random variable with probability density function given by

$$f(x) = \begin{cases} \frac{1}{2500}(10-x)^3 & 0 \leq x \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that $f(x)$ is a probability density function.

(b) What is the probability that between 3 and 5 thousand litres of petrol are sold in a week?

(c) If less than 2 thousand litres of petrol are sold in a week, the petrol is not supplied during the week that follows. What is the probability that the petrol company does not supply?

(d) What size tank is required in order to be 98% certain that the weekly demand can be met?

Solution 4.3.8. We are only going to show that $f(x)$ is a p.d.f then you are expected to complete the remaining questions.

(a) Once again, we required to show that $\int_a^b f(x)dx = 1$. That is,

$$\int_a^b f(x)dx = \frac{1}{2500}\int_0^{10} (10-x)^3 dx$$

Let $u = 10 - x$ so that $du = -dx$. Then we change the variable from x to u and have

$$\frac{1}{2500}\int (10-x)^3 dx = -\frac{1}{2500}\int u^3 du$$

Note the negative sign from $-dx$. This is much simpler to deal as its in polynomial form.

$$= -\frac{1}{2500}\left(\frac{u^4}{4}\right)$$

Substituting back $u = 10 - x$ we get

$$\begin{aligned}\frac{1}{2500}\int_0^{10} (10-x)^3 dx &= \left[-\frac{(10-x)^4}{10000}\right]_0^{10}\\ &= 1.\end{aligned}$$

Now that you have seen how to integrate this function, complete the rest of the questions.

4.4 Expectation and Variance of a Random Variable

In tt STA116, you learned about measures of location and spread for samples of data. We now develop equivalent concepts for random variables. The most important measures of location and spread for random variables are *expectation* (mean) and *variance*, as were used for samples of data. However, the formulas defining mean and variance of a random variable are completely different from the ones which defined the sample mean, $\bar{x}$ and sample variance, s^2. In addition, and as might have suspected, discrete and continuous random variables are handled separately. But this time around we keep the same notation.

4.4.1 Expectation

For a **discrete random variable** X with probability mass function $p(x)$, its expectation or the expected value of X, denoted by $\mathbb{E}(X)$ is given by

$$\mathbb{E}(X) = \sum_{x\,:\,p(x)>0} xp(x). \tag{4.7}$$

That is, the expected value is weighted average of the possible values that X can take on, each value being weighted by the probability that X assumes it.

For a **continuous random variable** X with probability density function, $f(x)$, its expected value is give by

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x)dx \tag{4.8}$$

Note that the limits of the integral are taken over the interval for which the probability is non-zero.

Example 4.4.1. Find the expected value of X, $\mathbb{E}(X)$ with the following probability mass functions.

(a)

$$p(x) = \begin{cases} \frac{1}{6} & x = 1, 2, 3, 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

(b)

$$p(x) = \begin{cases} \binom{3}{x} 0.2^x (0.8)^{3-x} & x = 0, 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

Solution 4.4.1. Solution will be provided during class.

Example 4.4.2. Consider Example 4.1.2. What is the expected value of the commission earned by the salesperson today?

Solution 4.4.2. Solution will be provided during class.

Example 4.4.3. Find the expected value of X, $\mathbb{E}(X)$ with the following probability density functions.

(a)

$$f(x) = \begin{cases} 6x(1-x) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(b)

$$f(x) = \begin{cases} 10x^9 & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Solution 4.4.3. Solution will be provided during class.

Example 4.4.4. A random variable X has probability density function

$$f(x) = \begin{cases} k(1-x^2) & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

Determine the value of k and hence find the expected value of X.

Solution 4.4.4. Solution will be provided during class.

Expectation of a Function

$$\text{Discrete:} \qquad \mathbb{E}\big[g(X)\big] = \sum_i g(x_i)p(x_i)$$

$$\text{Continuous:} \qquad \mathbb{E}\big[g(X)\big] = \int_{-\infty}^{\infty} g(x)p(x)dx$$

Example 4.4.5. Let X denote a random variable which takes on any of the following values $-1,\ 0,\ 1$ with the probabilities

$$p(-1) = 0.2,\ p(0) = 0.5\ p(1) = 0.3.$$

Compute $\mathbb{E}(X^2)$ and $\mathbb{E}(2X-1)$.

Example 4.4.6. A random variable X has probability density function

$$f(x) = \begin{cases} Ax + B & -1 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

If $\mathbb{E}(X) = 0.5$, find A and B then compute $\mathbb{E}(X^2)$.

Proposition 4.4.1. Let a,b be constants and X be a random variable. Then

$$\mathbb{E}\big(aX + b\big) = a\mathbb{E}\big(X\big) + b \tag{4.9}$$

Proof. We will only consider the discrete case but this can easily be extended to continuous random variables. Consider

$$\begin{aligned}\mathbb{E}\big(aX+b\big) &= \sum_x \big(ax+b\big)p(x)\\ &= \sum_x \Big[axp(x)+bp(x)\Big]\\ &= a\sum_x xp(x) + b\sum_x p(x)\\ &= a\mathbb{E}(X)+b\end{aligned}$$

4.4.2 Variance

The expected value of X yields the weighted average of the possible values of X and does not tell us anything about variation or spread of these values. For example, consider the following random variables with their respective probability mass functions:

$$W = 0 \qquad \text{with probability 1.}$$

$$Y = \begin{cases} -1 & \text{with probability } 1/2 \\ 1 & \text{with probability } 1/2. \end{cases}$$

$$Z = \begin{cases} -100 & \text{with probability } 1/2 \\ 100 & \text{with probability } 1/2. \end{cases}$$

All these have the same expectation of 0 but their spread are way different. Since we would usually expect X to take on values around its mean, $\mathbb{E}(X)$, it would appear that a reasonable way of measuring the possible variation of X would be to look at how far apart X would be from its mean, on average. We could consider $\mathbb{E}\Big(|X-\mu|\Big)$, where $\mu=\mathbb{E}(X)$, but this is mathematically inconvenient and therefore a more tractable quantity is considered.

Definition 4.4.1. If X is a random variable with mean (expected value), μ, then the variance of X, denoted by $\mathbb{V}(X)$ is defined by

$$\mathbb{V}(X) = \mathbb{E}\left[\left(X-\mu\right)^2\right] \tag{4.10}$$

Alternatively, the variance of X can be expressed as follows:

$$\begin{aligned}
\mathbb{V}(X) &= \mathbb{E}\left[\left(X-\mu\right)^2\right] \\
&= \mathbb{E}\left(X^2 - 2\mu X + \mu^2\right) \\
&= \mathbb{E}\big(X^2\big) - 2\mu\mathbb{E}\big(X\big) + \mu^2 \\
&= \mathbb{E}\big(X^2\big) - \mu^2
\end{aligned}$$

Thus,

$$\mathbb{V}(X) = \mathbb{E}\big(X^2\big) - \mu^2 \tag{4.11}$$

Exercise 1

Find the variance for all the questions discussed in the previous section.